Frozen Simulation Results

Two-class resource allocation and bounded capacity elasticity

Student: Fatima Subhi Khaleel
Research direction: SDN-managed O-RAN orchestration
Evaluation freeze: 18 October 2026

1. Scope and decision

The experiment is complete for the implemented abstraction. The results are not validated O-RAN network results and are not yet sufficient for a submission-ready Transactions paper.

This report freezes the code, comparator selection, raw outputs and interpretation. It does not rename a synthetic queueing experiment as a physical-layer or O-RAN protocol simulation.

2. Independent evaluation design

Element	Frozen choice
Calibration	Five seeds: 4000-4004; all 16 fixed transport/compute additions plus dynamic expansion
Evaluation	Twenty different seeds: 5000-5019; four settings and four comparison roles
Total new runs	340 calibration + 320 held-out
Measured cohort	300-slot warmup; 3000-slot arrivals; 40-slot followup; 20-ms synthetic slots
Static selection	Cheapest fixed addition with both calibration class means at or below 1%; fallback minimizes worst class mean
Budget comparator	Best calibration-weighted fixed allocation whose incremental cost is no greater than the calibration dynamic mean cost
Controller	19 class shares; bounded additions of 0-3 units per transport/compute stage; ten-slot actuation delay

The 1% value is a comparator-selection criterion on sample means. It is not a proven SLA, and comparator selection never uses held-out outcomes.

3. Held-out numerical results

Setting / method	Fixed addition	Class 0 (%)	Class 1 (%)	Extra cost
Nominal / Dynamic expansion	elastic_cap3	0.000	0.000	175.69
Nominal / Selected fixed expansion	static_0_0	0.000	0.000	0.00
Nominal / Budget-matched fixed	static_0_0	0.000	0.000	0.00
Transport stress / Dynamic expansion	elastic_cap3	0.000	0.070	3789.03
Transport stress / Selected fixed expansion	static_2_0	0.000	0.115	6680.00
Transport stress / Budget-matched fixed	static_1_0	0.517	16.452	3340.00
Compute stress / Dynamic expansion	elastic_cap3	0.000	0.015	3832.66
Compute stress / Selected fixed expansion	static_0_2	0.000	0.109	6680.00
Compute stress / Budget-matched fixed	static_0_1	0.551	14.326	3340.00
Delayed/lost reports / Dynamic expansion	elastic_cap3	0.000	0.058	4086.61
Delayed/lost reports / Selected fixed expansion	static_2_0	0.000	0.112	6680.00
Delayed/lost reports / Budget-matched fixed	static_1_0	0.889	16.540	3340.00

Cost includes incremental provisioned capacity and changes only, across warmup, measurement and followup. It excludes base infrastructure costs and uses abstract units, not currency.

4. Independent paired comparison

Setting	Dynamic - selected fixed, class 0 (pp)	Class 1 (pp)	Incremental cost saving
Nominal	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	Not defined: fixed cost is zero
Transport stress	+0.000 [+0.000, +0.000]	-0.044 [-0.133, +0.000]	43.28%
Compute stress	+0.000 [+0.000, +0.000]	-0.094 [-0.283, +0.000]	42.62%
Delayed/lost reports	+0.000 [+0.000, +0.000]	-0.054 [-0.329, +0.167]	38.82%

Brackets contain exploratory 95% paired seed-bootstrap intervals (2000 resamples). No multiplicity correction or non-inferiority test is applied. A zero bootstrap interval after zero observations does not establish zero true failure probability. Budget matching uses calibration mean cost, not a hard held-out cost equality.

5. Classwise performance

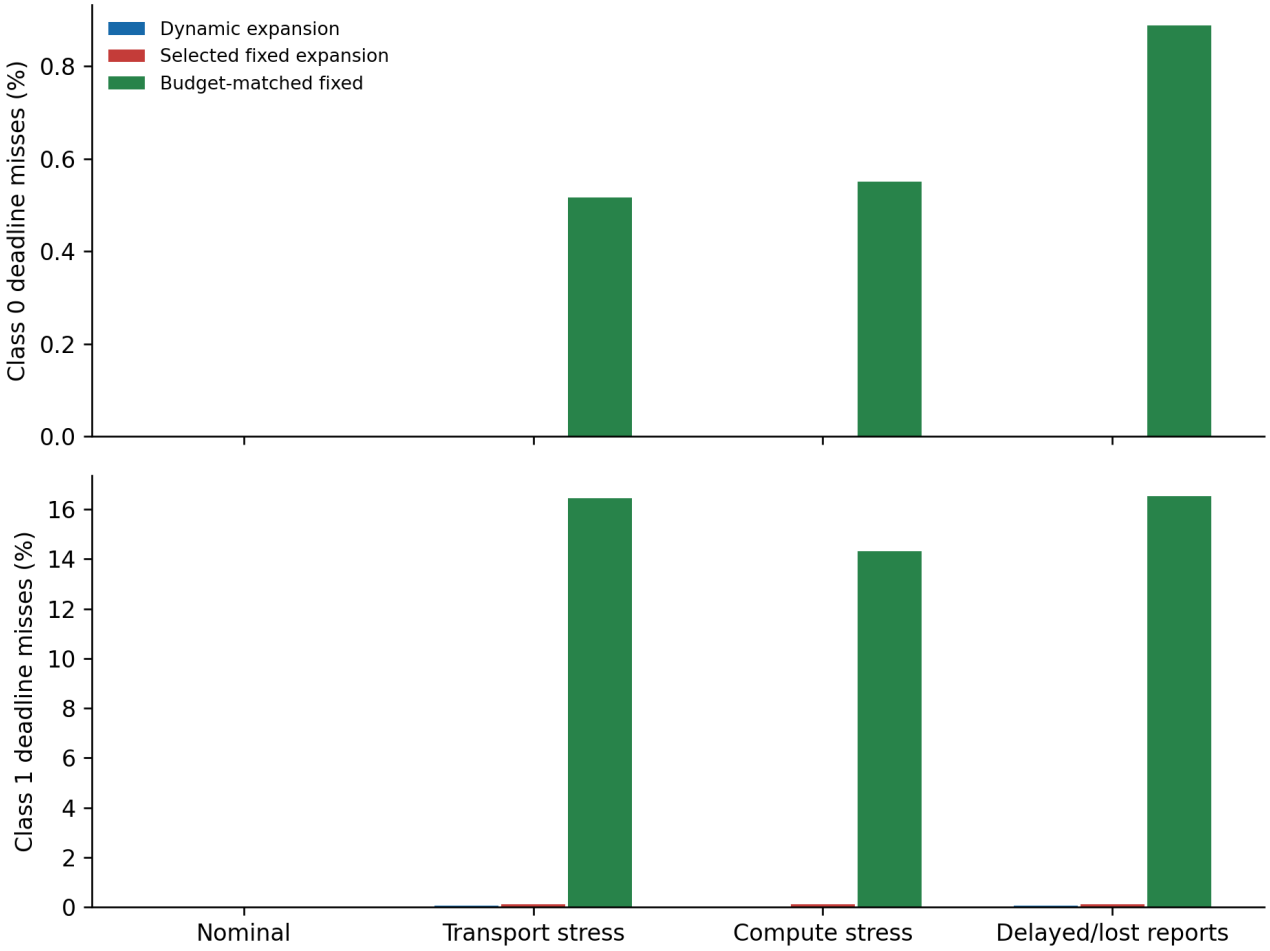


Figure 1. Both service classes are retained. Values are held-out seed means, with identical exogenous events and report request/loss budgets across comparison roles.

Do not infer that a zero-height bar is a formal reliability guarantee. The finite simulation can fail to observe rare events, and its synthetic workload does not cover real network distributions.

6. Resource cost

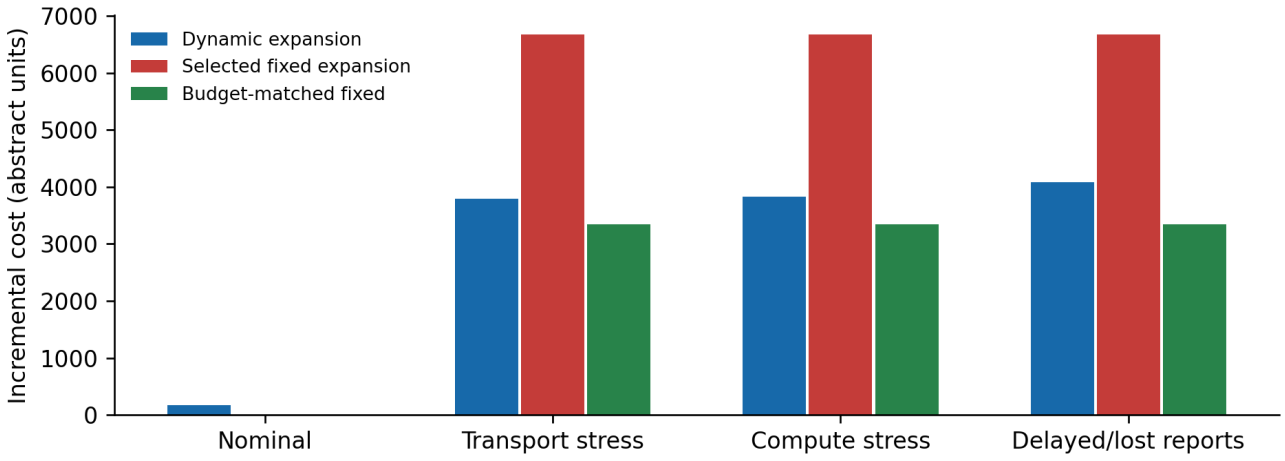


Figure 2. Incremental resource cost for the independently selected fixed comparator, budget-matched fixed comparator and dynamic policy. A lower cost must be considered alongside both classwise violation rates.

7. Interpretation of the earlier saving claim

The earlier 80-81% saving was measured against permanently maximum additions of three units at each expandable stage. It was not a saving against the cheapest adequate fixed allocation. This report replaces that weak comparison with a calibration-selected fixed baseline and independent evaluation. Any paper must use the stronger comparison and report adverse outcomes.

Nominal overhead

Fixed zero expansion can already meet all observed nominal deadlines. Any nonzero nominal dynamic expansion is additional cost without an observed quality benefit. It must be retained in the evaluation, not excluded as an inconvenient case.

8. Publication-readiness audit

Requirement	Current evidence / gap
Reproducible execution	Source, full per-seed output, independent selection, paired intervals, trace hashes and version-preserved baselines are supplied.
Internal verification	24 unit tests from the previous stages plus six static-engine numeric regression comparisons. Runtime conservation and capacity assertions execute in all runs.
Radio realism	No SINR, fading, interference, modulation/coding, retransmission or resource-block model. Radio is an abstract service queue.
Compute and transport	Packets require one service unit at each stage. No calibrated bit sizes, CPU cycles, propagation or telemetry bandwidth cost.
O-RAN integration	No E2/O1/O2 implementation, actual RIC deployment, SDN flow rules or cloud provisioner. Control split is conceptual.
Theoretical guarantees	Delay estimates and uncertainty margins are uncalibrated surrogates. Grid optimality is not optimality of the real network or a chance-constraint proof.
Literature and novelty	No systematic full-text closest-work comparison or same-journal evidence review is completed in this package.
External validation	No trace-calibrated workload, protocol emulator or real testbed cross-check.

Decision: retain these outputs as frozen results of the abstract model. Do not label them final publishable O-RAN results. Publication readiness requires a validated model and external comparisons, not merely more seeds or cosmetic figures.

9. Integrity and reproduction

Run frozen_evaluation.py to reproduce comparator selection and evaluation; run build_results_report.py to rebuild this report. Parameters and selection criteria are in the source. An integrity manifest records the input/output SHA-256 hashes. Raw files remain authoritative over rounded table values.